Integrated Management of Workloads and Energy System for Data Centers

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Abstract

Workload transfer is a critical and unique strategy for enhancing energy flexibility in data centers. However, existing research has largely focused on the operational cost benefits associated with workload transfer, neglecting its impact on the workloads themselves. To fully investigate the potential of workload transfer for sustainable and positive development in data centers, this study firstly clarifies the interaction between workload and energy. Based on this, an integrated management framework is proposed to simultaneously optimize workload transfer and energy dispatch. A transfer model that considers workload heterogeneity is developed to capture the characteristics of workload transfer and is integrated with the energy system model, enabling cross-sectoral collaboration between Information Technology (IT) and energy sectors. A case study in Shanghai reveals a clear trade-off exists between workload transfer and energy dispatch. The results show that while transferring workloads can reduce operational costs by 7.63% in summer and 2.81% in winter, it concurrently increases the average delay rate of workloads by 0.2434 in summer and 0.2028 in winter. The impact of workload transfer on energy systems is not always positive and a sole focus on cost reduction could result in an elevation of the average delay rate. Additionally, the driving forces behind workload transfer include variations in electricity prices, excess renewable energy and optimal operation of the energy system. The structure of workload also significantly affects the workload transfer. Overall, this study integrates modules from both the IT and energy sectors to achieve holistic management of data centers. The results of the case study provide novel and valuable insights for optimizing the operation of data centers during workload transfer processes.

Keywords: Data center, Workload transfer, Energy dispatch, Coupling relationship, Integrated management

Nomenclature

Abbreviations

ADR	average delay rate
AC	absorption chiller
CEEX	carbon emission cost
CPU	central processing unit
CS	cooling storage
DS	delay-sensitive
DT	delay-tolerant
EC	electricity cost
ED	Euclidean distance
FC	free cooling
GIE	gas internal engine
IT	internet technology
MC	maintenance cost
NG	natural gas
NC	fuel cost
OPEX	operation cost
PV	photovoltaic
QoS	Quality of Service
SRI	solar-radiation intensity
TOTX	total cost
UP	unit price

Indices

t	index for time slot
k	index for workload
j	index for electrical/cooling device

Variables

θ_k	delay rate of workload k
$D_{k,R}$	response time of workload k (hour)
$TOTX^t$	total cost at t^{th} time slot (CNY)
$OPEX^t$	operation cost at t^{th} time slot (CNY)
$CEEX^t$	carbon emission cost at t^{th} time slot (CNY)
NC^t , MC^t , EC^t	fuel/maintenance/electricity cost at t^{th} time slot (CNY)
NG_{GIE}^t	natural gas consumption at t^{th} time slot (kW)
$Q_{ele/cool,j}^t$	energy supply of electrical/cooling device j at t^{th} time slot (kW)
$\sigma_{ele,GIE}^t$	on/off status of GIE
$\sigma_{cool,AC}^t$	on/off status of AC
$\sigma_{cool,EC}^t$	on/off status of EC
$\psi_{ele,GIE}^t$	start limit variable of GIE
$Q_{cool,CS}^t$	cooling power of cooling storage tank at t^{th} time slot (kW)
$Q_{cool,CS,ch}^t$	cooling charging of CS at t^{th} time slot (kW)
$Q_{cool,CS,dis}^t$	cooling discharging of CS at t^{th} time slot (kW)
$\sigma_{cool,CS,dis}^t$	on/off status of cooling discharging
$\sigma_{cool,CS,ch}^t$	on/off status of cooling charging

Parameters

Ω	set of all workloads
N_Ω	total number of workloads in set Ω
$D_{k,d}$	time can be delayed of workload k (hour)
UP_{NG}	unit price of natural gas (CNY/kWh)
$UP_{MC,j}$	unit maintenance price of device j (CNY)
UP_{grid}^t	unit price of grid electricity at t^{th} time slot (CNY/ kWh)
UP_{CE}	unit price of carbon emission (CNY/kg)
π_{NG}	carbon emission factor of natural gas (kg/kWh)
π_{grid}	carbon emission factor of grid electricity (kg/kWh)
A^t	total workloads arriving at t^{th} time slot (requests)
DS^t	DS workloads arriving at t^{th} time slot (requests)
DT_a^t	DT workloads arriving at t^{th} time slot (requests)
$D_{k,c}$	computing time of workload k (hour)
μ	average processing speed of a server (requests/s)
m	number of servers
$t_{k,a}$	arriving time slot of workload k
Φ_k	set of time slots where workload k can be transferred to
D_{SC}	a single transfer cycle (hour)
ω	illumination coefficient (kW/m^2)
S_{lig}	illumination area (m^2)
P_{full} , P_{idle}	max/idle power of a server (kW)
$E_{cool,wa}^t$	heat transfer through envelope at t^{th} time slot (kW)
$E_{cool,lig}^t$	heat dissipation from lighting at t^{th} time slot (kW)
$E_{cool,pe}^t$	heat dissipation caused by individuals at t^{th} time slot (kW)
$Q_{ele,PV}^t$	electricity generation power of PV at t^{th} time slot (kW)
$\varphi_{ele,GIE}$, $\varphi_{heat,GIE}$	electricity/heat generation efficiency of GIE
S_{PV}	area of PV panels (m^2)
$\varphi_{ele,PV}$	electricity generation efficiency of PV
SRI^t	solar-radiation intensity at t^{th} time slot (W/m^2)

$Q^t_{ele,grid}$	electricity power from grid at t^{th} time slot (kW)	$\varphi_{cool,EC}$	coefficient of performance for EC
F^t	total workloads processed at t^{th} time slot (requests)	CAP_{EC}	cooling capacity of EC (kW)
DT_c^t	DT workloads processed at t^{th} time slot (requests)	CAP_{AC}	cooling capacity of AC (kW)
L^t	total workloads in the waiting queue at t^{th} time slot (requests)	$CAP_{ele,GIE}$	electricity generation capacity of GIE (kW)
$D_{k,q}$, $D_{k,w}$	queuing/waiting time of workload k (hour)	$\varphi_{cool,AC}$	coefficient of performance for AC
p^t_k	binary indicator of workload k being processed at t^{th} time slot	T^t_{TW}	transition temperature between full free cooling to partial free cooling (℃)
$t_{k,l}$	latest time slot by which processing of the workload k should begin	T^t_{sw}	supply cold water temperature (℃)
$t_{k,q}$	time slot of workload k starting queuing	T_{ap}	approaching temperature of the cooling tower (℃)
E^t_{ele}	total electrical demand at t^{th} time slot (kW)	T^t_{tr}	temperature rise of heat exchanger (℃)
$E^t_{ele,IT}$, $E^t_{ele,lig}$	electrical demand for IT/lighting equipment at t^{th} time slot (kW)	T^t_{rw}	return cold water temperature (℃)
$E^t_{ele,servers}$	electrical demand for servers at t^{th} time slot (kW)	M	mass flow rate (kg/h)
u^t	average utilization of a server (%)	T^t_{PW}	transition temperature between partial free cooling and mechanical cooling (℃)
E^t_{cool}	total cooling demand at t^{th} time slot (kW)	T^t_{wb}	outdoor wet bulb temperature (℃)
$E^t_{cool,IT}$	cooling demand for IT equipment at t^{th} time slot (kW)	$P^t_{ele,others}$	electrical demand for other auxiliary IT equipment at t^{th} time slot (kW)
$Q^t_{cool,EC}$	cooling supply of EC at t^{th} time slot (kW)	T^t_{ct}	output water temperature of cooling tower
$Q^t_{ele,EC}$	electricity consumption power of EC at t^{th} time slot (kW)	$\varphi_{cool,FC}$	efficiency of FC
$Q^t_{ele,GIE}$, $Q^t_{heat,GIE}$	electricity/heat generation power of GIE at t^{th} time slot (kW)	C	heat-specific capacity of water (kWh·kg^{-1}·℃$^{-1}$)
$Q^t_{cool,AC}$	cooling supply of AC at t^{th} time slot (kW)	CAP_{CS}	cooling capacity of CS (kW)
$Q^t_{cool,FC}$	cooling supply of FC at t^{th} time slot (kW)	$\varphi_{cool,CS,dis}$	efficiency of discharging
		$\varphi_{cool,CS,ch}$	efficiency of charging

1. Introduction

In pursuit of reducing carbon emissions and achieving carbon neutrality, every industry and sector faces significant challenges related to energy conservation, emission reduction, and energy transition. Data centers, the core carriers of infrastructure for the development of the digital economy, provide computility, which is considered a crucial strategic resource [1-2]. The rapid advancement of technologies such as Large Language Models, Blockchain, and the Internet of Vehicles is placing higher demands on computility, prompting a continuous growth in the number of data centers. As of 2023, the annual growth rate of China's data centers has consistently been around 30% for five years, with expectations of further increases in the years ahead [3]. Operating continuously throughout the year, data centers, as super energy-consuming sectors, consumed 270 TWh of electricity in 2021, accounting for approximately 2% of the global total electricity consumption [4]. This figure is anticipated to rise to 8% by 2030, potentially resulting in escalating carbon emissions [5]. Therefore, the economic and energy-efficient transition of data centers is an urgent issue that requires immediate attention to ensure healthy development of the data center industry.

In the face of high energy costs and high carbon emissions, data centers have implemented numerous energy-saving and low-carbon technologies. To reduce energy consumption of IT equipment, some scholars have studied start-stop optimization of servers, concentrating running workloads on some high-performance servers while putting inactive servers into idle mode, thus reducing power consumption [6]. In terms of reducing cooling power consumption, air-cooled data centers have conducted lots of studies on optimizing airflow organization to improve the utilization of cooling capacity [7-8]. More recently, there has been a growing interest in the local regulation of air flow, with the goal of achieving on-demand cooling to further reduce cooling power consumption [9-10]. In addition, the development of liquid cooling technology, such as immersion cooling [11], two-phase cooling [12] and spray cooling [13] has significantly reduced cooling power consumption. Research on the joint control of IT equipment and cooling systems, aimed at balancing the energy consumption of both to find the globally optimal solution, is also underway [14-15].

Among all those energy-saving and low-carbon technologies, workload transfer is unique to data centers. On the one hand, workload transfer relies on the communication network, enabling the energy demand generated by workload execution to be swiftly transferred without depending on the power grid [16]. On the other hand, a data center is an integrator of multiple sectors, with the energy sector and IT sector playing predominant roles. The energy sector is responsible for providing power, while the IT sector ensures functionality. This makes the energy dispatch of the energy sector and the workload transfer of the IT sector the primary focus of data centers. Variations in workload transfer can regulate the energy demand and fully exploit the flexibility of the demand side, enabling the supply side to provide more flexible low-carbon energy supply. However, despite being a combination of energy and IT sectors, data centers currently manage energy sector (supply side) and IT sector (demand side) separately, as well as analyze workload transfer and energy dispatch separately. This leads to significant potential for economic and environmental optimization.

Traditional data centers typically rely on a single energy source and lack refined management of supply-side operations [17-18]. Faced with the challenges of substantial electricity usage and carbon emissions, data centers have initiated efforts to improve their energy system structure, primarily through the following four approaches: using renewable energy [19-20]; employing more efficient cooling methods [21]; implementing cascade energy utilization [22-23], using energy storage devices [24-25] and performing economic analyses of energy systems [26-27].

The use of renewable energy is one of the most important strategies in data centers. It is a common practice to plan and install solar panels within a data center park. For instance, the *Tencent* Data Center in China has mounted solar panels on its building roof with a capacity of 10.54 MW. This setup can generate up to 12 million kWh annual electricity, greatly reducing carbon emissions [28]. The data center in Inzai, Japan used wind power to increase the flexibility of its battery energy storage system, thereby reducing battery investment costs [19]. Free cooling is an efficient cooling method and should be fully considered in the construction of data centers, as mandated by GB50174-2017 [29]. Innovative cooling systems for data

centers, incorporating indirect free cooling and water-side economizers, have been proposed to cut down the energy consumption of cooling systems by as much as 15% [30]. Additionally, some data centers are strategically located adjacent to lakes. By using the consistent and low temperature of lake water, a water source free cooling system has been deployed, securing a more stable and extended supply of free cooling [31].

The adoption of cascade energy utilization in data centers significantly enhances their energy efficiency. For instance, a novel system integrating a proton exchange membrane fuel cell for combined cooling and power, equipped with multiple cooling devices, has been developed for data centers, increasing the annual energy efficiencies to 86.53% [32]. Additionally, an adsorption refrigeration system has been proposed for the recovery of very low temperature waste heat (up to 60°C) from data centers, demonstrating its economic advantage [33]. Another innovative approach combines CO_2 ground source heat pump waste heat recovery with soil heat storage for building heating, resulting in a reduction of total energy consumption to 52.86% of its initial level [34].

Energy storage plays a vital role in data center operations. Cooling storage, often used for emergency responses in disaster scenarios [29], has also been applied to boost energy efficiency [35]. Researcher have also conducted studies on the use building's physical characteristics for cooling storage, proposing to overcool the data center in early morning to store part of the cooling inside building to reduce energy cost [36]. Similarly, electrical storage system, while primarily serving as disaster prevention measures in data centers, are increasingly utilized for power load regulation [24].

In addition to developing sustainable energy systems, conducting economic analyses of data center is also vital. A synergistic energy dispatch approach integrating hydrogen technology, fuel cells and electric vehicles has been proposed for energy management of data centers and their adjacent buildings, reducing operation cost and carbon emission by 8.2% and 3.9%, respectively [37]. Researchers compared economic efficiency of air source heat pump tobacco drying and data center waste heat tobacco drying and found that the latter one can reduce investment by 21% [38]. However, most of the above-mentioned studies are based on a

given set of data center workloads and do noy take into account of the transfer of workloads, thereby not fully developing the potential of the data center's demand side.

As a distinctive method of regulating demand-side operations, workload transfer has proven to be an effective strategy for optimizing data center operations. In the field of computer science, workload scheduling is one of the key research fields. In fact, workload scheduling not only affects the energy consumption of IT equipment but also has a significant impact on the energy usage of cooling systems. Power-aware scheduling can significantly reduce server power consumption [39]. This can be achieved through various strategies, such as optimizing the placement of workload or virtual machines [40-41], as well as refining the timing of server activation and deactivation [6]. Thermal-aware scheduling is also an important research branch. It primarily includes two approaches: proactive and reactive scheduling [42]. Optimization can be carried out from two perspectives, server relocation [43-44] and workload scheduling [14, 45], with the goal of reducing server outlet temperatures and thereby lowering the energy consumption of the cooling system. In addition, some studies have examined the thermal response of servers under different workload levels to identify and avoid thermal hotspots [46]. Other study has introduced the concept of thermal stress analysis, using inlet temperature effect to monitor server status and guide server migration, which helps improve cooling efficiency and hardware stability [47]. These studies, focusing on server-level workload scheduling, have significantly reduced either IT equipment energy consumption or cooling energy consumption. From the perspective of our research focus, optimizing workload transfer and energy dispatch is also a vital and active area of research. The development of research in this area can be broadly categorized into three stages. Initially, the focus was on reducing operational costs of data centers by workload transfer. Related research can be traced back to 2009, when Qureshi A first proposed using electricity price differences to reduce data center operating costs by transferring workloads to lower-priced times [48]. Subsequently, the idea of using workload transfer as a demand response method has attracted widespread attention in the academic community [49]. It is reported that in large-scale data centers, fluctuations in their demand can impact electricity market prices. Some scholars have studied the dynamic electricity pricing in

relation to workload transfer in the data center, and proposed a price-sensitivity workload transfer strategy, which effectively reduces operational costs for data center operators [50].

In the second stage, the scope of research broadened to encompass not just cost savings but also the integration of workload transfer with additional technologies. In terms of workload transfer and IT system, some scholars have studied the relationship between workload transfer and the optimization of data center servers' startups. They proposed adjusting the number of active servers through workload transfer, thereby reducing the energy consumption of the IT system [6]. Additionally, there has been research on the joint control of workload transfer and dynamic voltage and frequency scaling, suggesting that reducing server frequency during low workloads period can save a significant percentage of total energy consumption [51]. In terms of workload transfer and cooling systems, scholars have investigated the impact of workload management on the data center cooling systems. By optimizing control parameters of the cooling system while ensuring successful workload execution after workload transfer, it is possible to enhance the coefficient of performance of the cooling system by up to 20% [14].

In the third stage, workload transfer provides opportunities for optimizing the energy systems of the data center. Scholars have proposed a two-level optimization model, where the upper-level model is used to optimize the power supply cost of the generator set, and the lower level minimizes the difference between actual output and planned output through workload transfer, thereby reducing the overall operating cost of the power system [52]. The workload transfer mentioned above is mostly a temporal transfer, but there has been significant research into spatial transfer as well. Researchers have studied the economic benefits of spatial workload transfer from the East to the West in the United States, which could utilize 62% of the renewable energy curtailment and potentially achieve negative emission reduction costs [53]. Furthermore, global-scale workload transfer has also been studied, utilizing the complementary nature of renewable resources across different locations. Studies on such workload transfer across four data centers located in Georgia, Oregon, Dublin, and Zhangbin has shown a reduction in total carbon emissions by 47.38%, helping to achieve carbon neutrality in large-scale deployment of data centers worldwide [54].

From the aforementioned studies, we can see that current research mostly focuses on the energy sector, exploring the economic and environmental benefits achievable through workload transfer. These studies typically consider only the unidirectional impacts between the IT and energy sectors, particularly neglecting the impact of energy system operation on the workloads. In contrast, this paper proposes a more comprehensive strategy that collaboratively optimizes workload transfer performance and energy system costs within a single data center. Considering the heterogeneity of workloads, a workload transfer model is constructed and integrated with the energy system model. The proposed integrated optimization framework couples the supply-side and demand-side modules, facilitating bidirectional interactions between workload transfer and energy dispatch. This approach not only promotes the sustainable development of both energy and IT sectors but also enhances cross-sector collaboration, which is crucial for the sustainable development of data centers. The main contributions of this paper are as follows.

(1) The coupling relationship between energy and data is revealed through a thorough analysis of workload execution on the workload side and energy supply on the energy system side.

(2) A model for transferring workloads that accounts for heterogeneity is established, alongside a model for the interaction between workload transfer and energy dispatch to capture the coupling relationship between energy and data.

(3) A co-optimization framework integrating the IT and energy sectors is proposed, with the goal of optimizing energy costs and average delay rates. The mutual impact of workload transfer and energy dispatch are thoroughly explored.

The remaining sections of this paper are organized as follows: Section 2 reveals coupling relationship between energy and data in data centers, addressing both the energy and workload aspects. Section 3 describes the methodology, including the framework and models. Section 4 introduces a case study to demonstrate the application of the framework and models. Section 5 presents an analysis and discussion of the results. Section 6 summarizes the main conclusions.

2. Coupling relationship between workload transfer and energy dispatch

In data centers, there are two main ways of workload scheduling: centralized and decentralized. In centralized workload scheduling, workloads submitted by users are collected by the front-end servers and then allocated to various data centers. The front-end server acts as the decision maker, and all data centers share information with it to determine the allocation of workloads to specific data centers. In decentralized workload scheduling, there is no front-end server. Instead, peer-to-peer transactions are utilized, with the blockchain serving as the system framework. All transactions are broadcasted to all data centers for verification, thereby eliminating the necessity of a front-end server [55]. Workloads distributed by the front-end servers or through peer-to-peer transactions, upon arriving at the data center, will have their execution time and server nodes determined by the internal energy management strategy of the single data center. The final results will be returned to the user. The above process is the basic workflow for workload execution. Concurrently, the data center's energy system supplies cooling and electrical energy to support the data center's fundamental operational requirements. Interaction occurs between two key sectors: the IT sector, which handles workload transfer, and the energy sector, responsible for operating the energy system. This section will detail the coupling relationship between these two sectors in a single data center, as shown in Figure 1.

2.1. Workload transfer side

Data in this context refers to the workloads awaiting execution in data centers, which can be categorized into two main types: delay-sensitive workloads and delay-tolerant workloads [56]. Delay-sensitive workloads, often with strict deadlines measured in milliseconds or seconds [49], require immediate processing upon arrival at the data center. Consequently, these workloads cannot be transferred across time. In contrast, delay-tolerant workloads, which have much longer deadlines ranging from minutes to hours, can be transferred flexibly within the time domain as long as their deadlines are met. The flexibility of delay-tolerant workloads to be time-transferred forms the foundation of workload management in data centers. By transferring these workloads, the time sequence of the data center's loads can be altered.

The processing of workloads requires computing resources, such as Central Processing Unit (CPU), memory and disk, from IT equipment. The volume of workloads to be processed typically determines the required amount of computing resources. As the time sequence of workloads varies, the demand for computing resources in the data center adapts accordingly to guarantee the timely completion of all workloads.

On the workload transfer side, transferring workloads changes the time sequence of loads in the data center, consequently reshaping the computing demand curve of IT equipment.

2.2. Energy dispatch side

The energy consumption of IT equipment in data centers is directly correlates computing demand. As the demand for computing resources increases, the data center either activates additional servers or increases resource utilization of existing servers to ensure workloads completion, resulting in higher total electrical demand for IT equipment. The electrical consumption by IT equipment generates heat, and higher electrical consumption leads to higher heat generation, thereby increasing the cooling demand of the data center. In essence, fluctuations in the demand for computing resources dictate the overall energy demand of the data center.

Variations in cooling and electrical demands significantly influence the energy supply strategy of data center energy systems. Traditionally, data centers depended exclusively on grid-imported power. However, modern data centers are adopting a multi-energy strategy for their energy supply and displaying characteristics of integrated multi-energy coupling within their energy systems. Consequently, in response to fluctuating demand, a data center's energy system utilize a variety of energy supply strategies. The output of different energy equipment is adjusted accordingly to ensure that operational costs and carbon emissions are minimized while fulfilling the energy needs of the data center.

The output of the energy equipment directly influences the scheduling of workloads in the data center. Essentially, the transfer of workloads is not random but is limited by the output capabilities and operational constrains of the energy system.

On the energy dispatch, the fluctuating demand for computing resources determines the energy demand of the data center, which in turn affects the energy supply strategy of the energy system. Finally, this strategy of energy supply significantly influences the transfer of workloads in the data center.

In summary, a bidirectional coupling exists between workload transfer and energy dispatch within a data center. This interaction significantly impacts the economic and environmental performance of the energy system, as well as the performance of workload execution.

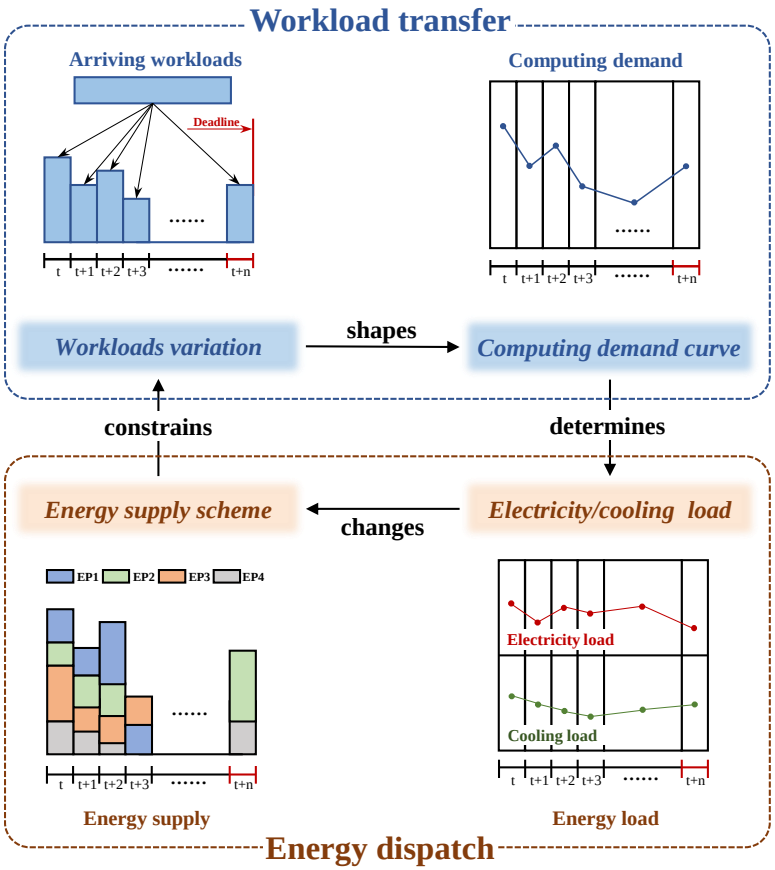


Figure 1. Schematic diagram of the relationship between workload transfer and energy dispatch.

3. Methodology

3.1. Intergrated management framework of workload transfer and energy dispatch

In current literature, workload transfer is primarily viewed as a method for regulating operations of energy systems, providing adjustable time intervals for managing the energy

demand of data centers and opportunities to optimizing the operation costs of energy systems. However, as discussed above, there is a bidirectional coupling relationship between workload temporal transfer and energy system dispatch. The bidirectional interaction needs to be studied to achieve cross-sectoral optimization between IT and energy sectors. The framework proposed to address this need is presented in this section.

3.1.1. Objective functions

Firstly, the objective of the workloads side is defined. A schematic diagram, shown in Figure 2, illustrates the entire life cycle of a workload, including transmission time, waiting time, queuing time, calculation time, and another transmission time. In this diagram, the transmission time denotes the period linked to the cross-regional scheduling of the workload, which falls outside the scope of this paper. The diagram's central portion focuses on the processing of workloads within the data center, a duration also referred to as the response time of the workload.

Upon arrival at the data center, the delay-sensitive workload is directly allocated to the computing servers, while the delay-tolerant workload is initially placed in the waiting queue, where it experiences a waiting time. It is then assigned to the computing server's queue, entering a period known as the queuing time, during which it waits for processing. Finally, the workload is moved to the appropriate hardware for computation, marking the beginning of the computing time, which continues until the workload's completion.

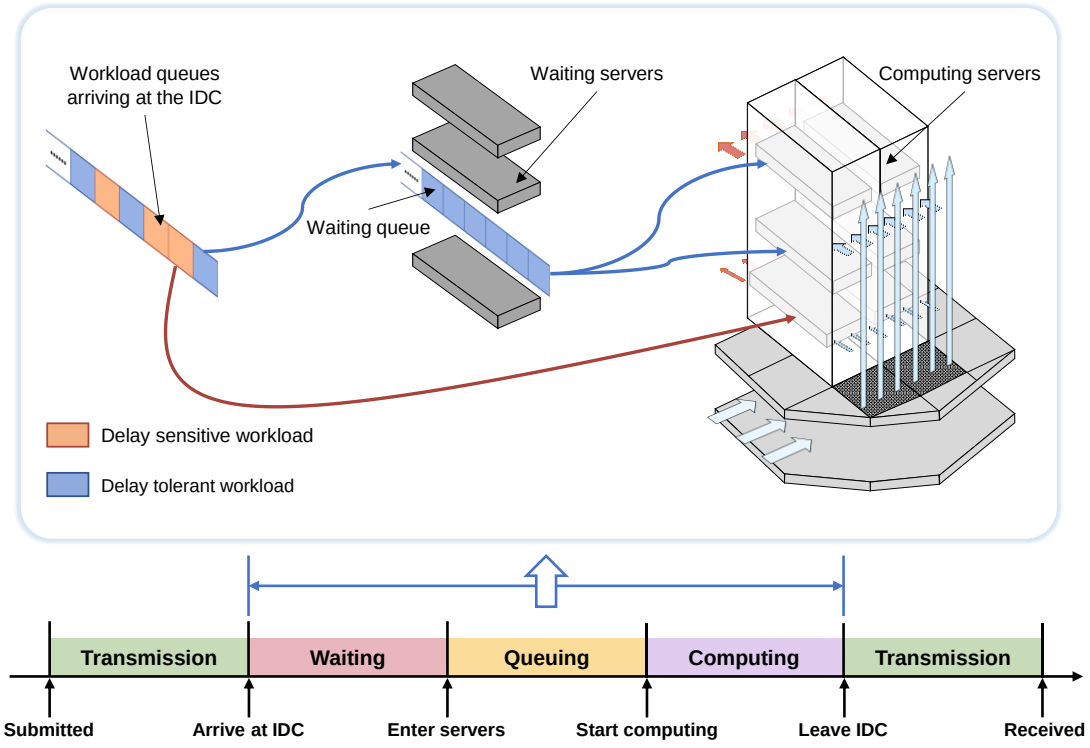


Figure 2. Schematic diagram of task load execution process

In data center development, Quality of Service (QoS) plays a crucial role [57]. QoS has a direct influence on user satisfaction, with user "|happiness" declining sharply if it falls below their minimum expectations. Conversely, their "happiness" increases within the bounds of expected QoS levels as QoS improves. QoS not only affects users' trust and their positive experience with the data center but also influences users' likelihood of future engagements, thereby impacting user retention and potentially leading to decreased customer loyalty if not managed well. This, in turn, can negatively impact the data center's long-term growth and sustainability. Moreover, the operational revenue of data centers is tightly linked to QoS. According to the literature [58], higher QoS allows data centers to charge more for their services. This connection highlights the dual importance of high QoS in ensuring both customer satisfaction and the financial well-being of the data center.

The response time of workload is the most utilized QoS metric in the field of service computing [57]. There is an inverse relationship between QoS and response time, meaning that shorter response time brings higher QoS levels. By minimizing response time, data centers can

enhance their revenue and promote healthy development. In light of this, the average delay rate (ADR) of workloads has been identified as an optimization target on the workload side, as shown in Eq. (1) and (2). Addressing this optimization challenge by minimizing the ADR for all workloads contributes to enhancing the overall QoS of the data center.

$$ADR = \frac{\sum_{k\in\Omega} \theta_k}{N_\Omega} \tag{1}$$

$$\theta_k = \frac{D_{k,R}}{D_{k,d}} \tag{2}$$

Where θ_k is delay rate of workload *k*. It is the ratio of response time $D_{k,R}$ to delayable time $D_{k,d}$. N_Ω is the total number of workloads and Ω is the set of all workloads.

On the energy side, its total cost (*TOTX*) is selected as the objective. *TOTX* is divided into two parts: operation cost (*OPEX*) and carbon emission cost (*CEEX*). *OPEX* is the sum of fuel cost (*NC*), maintenance cost (*MC*) and electricity cost (*EC*). *CEEX* is calculated by multiplying the total carbon emissions by a carbon tax rate. The equations are listed in Eq.(3) - (8).

$$TOTX^t = \sum_{\Psi} \left(OPEX^t + CEEX^t \right) \tag{3}$$

$$OPEX^t = NC^t + MC^t + EC^t \tag{4}$$

$$NC^t = NG_{GIE}^t \times UP_{NG} \tag{5}$$

$$MC^t = \sum_j (Q_{ele/cool,j}^t \times UP_{MC,j}) \tag{6}$$

$$EC^t = Q_{ele,grid}^t \times UP_{grid}^t \tag{7}$$

$$CEEX^t = \left(\pi_{NG} \times NG_{GIE}^t + \pi_{grid} \times Q_{ele,grid}^t \right) \times UP_{CE} \tag{8}$$

3.1.2. Integrated management framework

The integrated management framework, as depicted in Figure 3, uses Pareto optimal solutions to address two conflicting objectives. This study applies ε-constraint method to obtain the Pareto frontier [59], where each solution represents a non-dominated solution within the energy-data optimization problem. This means that within the feasible domain, no other solution can simultaneously improve the objective function values for both energy and workload. After obtaining the Pareto solution set, technique for order preference by similarity

to ideal solution (TOPSIS) is used for decision-making. This approach identifies the optimal solution based on its comprehensive distance. The comprehensive distance is characterized as the Euclidean distance from a non-dominated solution to the non-ideal point, surpassing the combined Euclidean distances from the solution to both the ideal and non-ideal points, as illustrated in Eq.(9) - (12).

$$i_{op} = \arg\max Y_i \tag{9}$$

$$Y_i = \frac{ED_{i-}}{ED_{i+} + ED_{i-}} \tag{10}$$

$$ED_{i+} = \sqrt{\sum_{j=1}^{n} \left(f_{ij} - f_j^{Ideal} \right)^2} \tag{11}$$

$$ED_{i-} = \sqrt{\sum_{j=1}^{n} \left(f_{ij} - f_j^{Non-ideal} \right)^2} \tag{12}$$

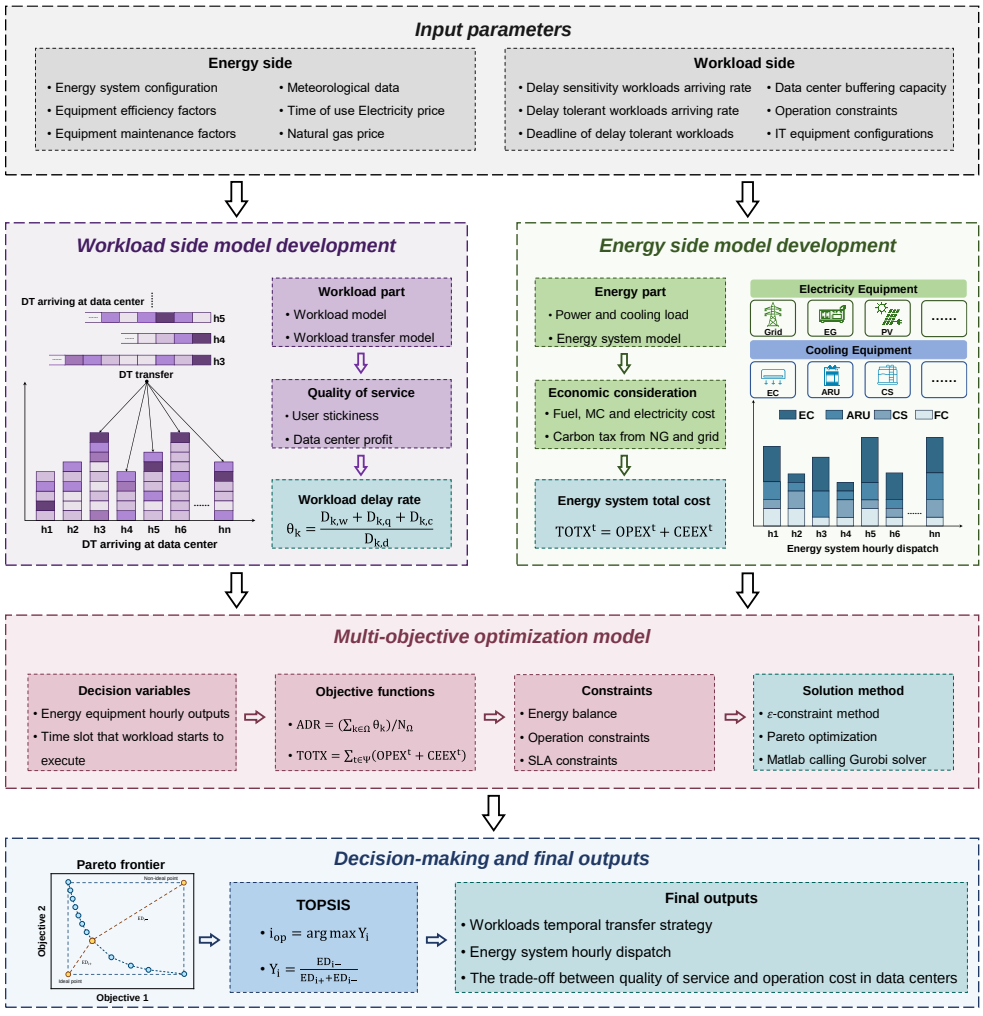


Figure 3. Schematic diagram of the integrated management framework.

3.2. Description of workload side

In order to manage the data center effectively, this section will describe the execution and scheduling of the workloads in the data center.

3.2.1. Workload model

A^t is the total workloads arriving at the data center at the tth time slot, including delay-sensitive workloads DS^t and delay-tolerant workloads DT_a^t, shown in Eq. (13).

$$A^t = DS^t + DT_a^t \tag{13}$$

DS^t are executed immediately, whereas some DT_a^t might be queued in the waiting queue. F^t denotes the total workloads processed at the tth time slot, which equals to DS^t plus DT_c^t (the total delay-tolerant workloads executed at t^{th} time slot), as shown in Eq. (14).

$$F^t = DS^t + DT_c^t \tag{14}$$

The workloads in the waiting queue L^t is obtained according to Eq. (15). L^{t-1} is the length of waiting queue at last time slot. Workloads are assumed to be executed under the condition that all dependency constraints have been met [24, 60].

$$L^t = L^{t-1} + A^t - F^t \tag{15}$$

$D_{k,R}$ is the response time of workload *k*, given by Eq. (16). According to the M/M/1 queuing theory [49], the average queuing time $D_{k,q}$ and average computing time $D_{k,c}$ of the workload in the computing server are calculated by Eq. (17) and (18). Waiting time $D_{k,w}$ is decided by workload transfer and is introduced in the next section.

$$D_{k,R} = D_{k,q} + D_{k,c} + D_{k,w} \tag{16}$$

$$D_{k,q} = 1/\left(\mu - \frac{F^t}{m}\right) \tag{17}$$

$$D_{k,c} = \frac{1}{\mu} \tag{18}$$

Where m is the number of active servers. In this study, the start-stop optimization of servers is not taken into consideration, so the number of active servers is a constant. μ is average processing speed of the server.

3.2.2. Workload transfer model

For delay-sensitive workloads, due to their strict deadline requirements, they are executed immediately upon arrival at the data center. Consequently, there is no workload transfer for them.

For delay-tolerant workloads arriving at t^{th} time slot, after they arrive at the data center, they do not need to be executed immediately, and their execution time is limited by the deadlines. The workload transfer of DT_a^t is presented in Eq.(19) which represents the workload transferred out of tth time slot.

$$DT_a^t = \sum_{\tau \in \Phi_k} \sum_{k \in DT_A^t} p_k^\tau, \Phi_k = \left\{ t, t+1..t_{k,l} \right\} \tag{19}$$

$$t_{k,l} = t_{k,a} + \left\lfloor D_{k,d} - \left(D_{k,q} + D_{k,c} \right) \right\rfloor \tag{20}$$

$$\sum_{\tau \in \Phi_k} p_k^\tau = 1 \tag{21}$$

Where $p_k^{t+1} = 1$ if workload k is transferred to $(t+1)^{th}$, otherwise $p_k^{t+1} = 0$. Φ_k^t is the set of time slots to which workload k can be transferred. This study assumes that each delay-tolerant workload can be transferred a maximum of one time, and that it experiences no workload execution failures, interruptions or restarts, so the constraint of Eq.(21) is added. $t_{k,l}$ is the latest time slot by which processing of the workload begins.

In the tth time slot, the total quantity of delay-tolerant workloads executed by data center is calculated by Eq.(22). It represents the workload transferred to the tth time slot. All delay-tolerant workloads must be completed within a single transfer cycle D_{sc}, thus fulfilling the constraint of Eq.(23), in this study D_{sc} is 24 hours.

$$DT_c^t = \sum_{\tau=1}^{t} \sum_{k \in DT_A^\tau} p_k^t \tag{22}$$

$$\sum_{t=1}^{D_{sc}} DT_c^t = \sum_{t=1}^{D_{sc}} DT_a^t \tag{23}$$

Finally, waiting time of workload k is calculated by Eq.(24), and the response time of workload k can be obtained combining Eq.(16) - (18).

$$D_{k,w} = t_{k,q} - t_{k,a} = \tau - t\big|_{k \in DT_A^t, p_k^\tau = 1} \tag{24}$$

Where $t_{k,q}$ is the time when workload *k* enters the computing server and $t_{k,a}$ is the time when workload *k* arrives data center.

3.3. Description of energy side

Generally, data center energy system includes microgrid, cooling network, and heating network. The heating network primarily serves to provide heating to office buildings during the winter months. Since the demand for heating is significantly lower than that for cooling, this study will not investigate the energy supply to office buildings. The subsequent sections will detail the energy side of data center.

3.3.1. Electrical and cooling demand

The electrical demand could be calculated by Eq.(25).

$$E_{ele}^{t} = E_{ele,IT}^{t} + E_{ele,lig}^{t} \tag{25}$$

Where $E_{ele,lig}^{t}$ is electrical demand for lighting equipment which can be obtained by Eq. (26). $E_{ele,IT}^{t}$ is electrical demand for IT equipment which consists of two parts: electrical demand for servers and electrical demand for other IT auxiliary equipment, as shown in Eq. (27). A widely used server utilization-based model [61-62] is adopted to calculate the electrical demand for servers, as shown in Eq. (28). The demand of other auxiliary IT equipment ($E_{ele,\text{others}}^{t}$) is set as a constant value in this study [63]. u^t is average utilization of a server and it is calculated by Eq. (29) [24, 60].

$$E_{ele,lig}^{t} = \omega \times S_{lig} \tag{26}$$

$$E_{ele,IT}^{t} = E_{ele,servers}^{t} + E_{ele,others}^{t} \tag{27}$$

$$E_{ele,servers}^{t} = m \times \left[u^t \times \left(P_{full} - P_{idle} \right) + P_{idle} \right] \tag{28}$$

$$u^t = F^t / m\mu \tag{29}$$

Where ω is illumination coefficient and S_{lig} is illumination area. m is the number of active servers. P_{full} and P_{idle} are maximum power and idle power of a server respectively. The maximum power refers to the energy consumed when the server operates at full capacity, while

the idle power denotes the energy used when the server is on but not actively processing data, maintaining only the essential functions of the server system.

The cooling demand can be calculated by Eq. (30)

$$E_{cool}^{t} = E_{cool,wa}^{t} + E_{cool,IT}^{t} + E_{cool,lig}^{t} + E_{cool,pe}^{t} \tag{30}$$

Where $E_{cool,wa}^{t}$ is the heat transfer through the building envelope. $E_{cool,IT}^{t}$ is the heat dissipation of the IT equipment. $E_{cool,lig}^{t}$ represents heat dissipation from lighting, while $E_{cool,pe}^{t}$ is heat dissipation caused by individuals, which is extremely small in data centers and is thus ignored in this study.

3.3.2. Energy system model

Figure 4 presents the schematic diagram of the energy system in this study, which includes a microgrid and a cooling network. Description of these components are detailed in the follow section. It should be noted that this study focuses on the integrated management of workload transfer and energy dispatch. Therefore, this study focuses on the impact of workload transfer on data centers at the energy system level. A detailed model of energy system operation is established, while server-level differences in power consumption, heat generation, and cooling characteristics are not incorporated in this study.

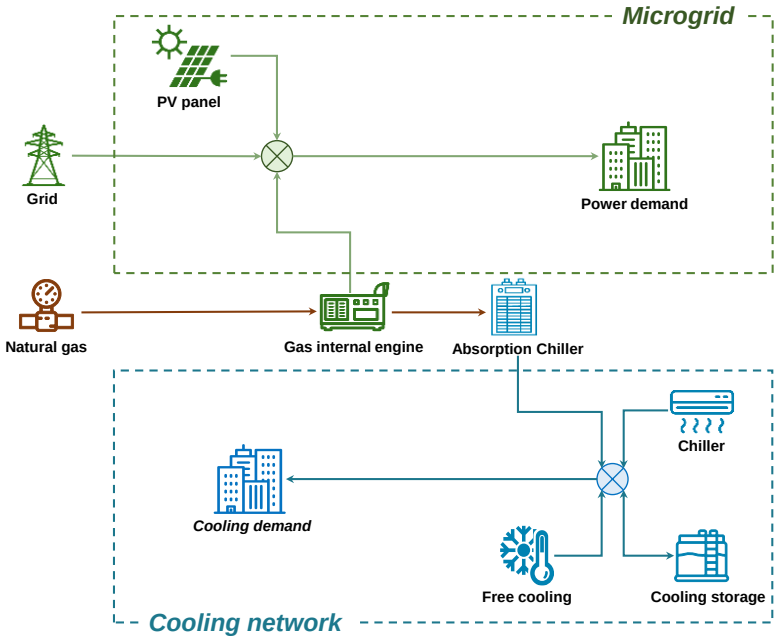


Figure 4. Schematic diagram of the energy system

The energy balance of the microgrid is shown in Eq. (31).

$$E_{ele}^{t} = Q_{ele,grid}^{t} + Q_{ele,GIE}^{t} + Q_{ele,PV}^{t} - Q_{ele,EC}^{t} \tag{31}$$

Where $Q_{ele,grid}^{t}$ is power import from grid. $Q_{ele,GIE}^{t}$ is power output of gas internal engine and calculated by Eq. (32). $Q_{heat,chp}^{t}$ is heat output of gas internal engine calculated by Eq. (33).

$$Q_{ele,GIE}^{t} = \varphi_{ele,GIE} \times NG_{GIE}^{t} \tag{32}$$

$$Q_{heat,GIE}^{t} = \varphi_{heat,GIE} \times NG_{GIE}^{t} \tag{33}$$

Moreover, the gas internal engine is subject to specific operational constraints. Eq.(34) and (35) guarantee that the output stays below its capacity while exceeding 15% of capacity [64], maintaining operation at optimal energy efficiency. Eq. (36) imposes a ramping constraint on the unit to avoid abrupt fluctuations in output, guaranteeing that any change in output between two time slots stays within its ramping capability. Eq.(37)-(39) is the start-stop constraints of GIE, which ensure that the operation is started at most once per day [65].

$$Q_{ele,GIE}^{t} \le \sigma_{ele,GIE}^{t} \times CAP_{ele,GIE} \tag{34}$$

$$Q_{ele,GIE}^{t} \ge 15\% \times \sigma_{ele,GIE}^{t} \times CAP_{ele,GIE} \tag{35}$$

$$\left| Q_{ele,GIE}^{t} - Q_{ele,GIE}^{t-1} \right| \le 50\% \times CAP_{ele,GIE} \tag{36}$$

$$\sum_{t=1}^{D_{sc}} \psi_{ele,GIE}^{t} = 1 \tag{37}$$

$$\sigma_{ele,GIE}^{t} - \sigma_{ele,GIE}^{t-1} \le \psi_{ele,GIE}^{t} \le 1 - \sigma_{ele,GIE}^{t-1} \tag{38}$$

$$\psi_{ele,GIE}^{t} \le \sigma_{ele,GIE}^{t} \tag{39}$$

The output of PV can be calculated by Eq. (40) [66].

$$Q_{ele,PV}^{t} = S_{PV} \times \varphi_{ele,PV} \times SRI^{t} \tag{40}$$

Where $\varphi_{ele,PV}^{t}$ is the efficiency of PV power generation. S_{PV} is the area of PV panels. SRI^{t} is solar radiation intensity.

The energy balance of the cooling network is shown in Eq. (41).

$$E_{cool}^{t} = Q_{cool,EC}^{t} + Q_{cool,AC}^{t} + Q_{cool,FC}^{t} + Q_{cool,CS,dis}^{t} - Q_{cool,CS,ch}^{t} \tag{41}$$

$Q^t_{cool,EC}$ is the cooling supply of the electric chillers, and it is constrained by Eq. (42) and (43).

$Q^t_{cool,AC}$ is the cooling supply of the absorption chillers, and it is calculated by Eq. (44) and (45).

$$Q^t_{cool,EC} \le \sigma^t_{cool,EC} \times \varphi_{cool,EC} \times Q^t_{ele,EC} \tag{42}$$

$$Q^t_{cool,EC} \le \sigma^t_{cool,EC} \times CAP_{EC} \tag{43}$$

$$Q^t_{cool,AC} = \sigma^t_{cool,AC} \times \varphi_{cool,AC} \times Q^t_{heat,GIE} \tag{44}$$

$$Q^t_{cool,AC} \le \sigma^t_{cool,AC} \times CAP_{AC} \tag{45}$$

$Q^t_{cool,FC}$ denotes the cooling supply provided by free cooling, which is influenced by the outdoor temperature. Depending on the outdoor wet bulb temperature, free cooling can be categorized into three modes: full free cooling, partial free cooling and mechanical cooling, as shown in Eq. (46) – (50) [67]. These modes are differentiated by specific threshold temperatures: T^t_{TW} represents the transition temperature from full free cooling to partial free cooling; T^t_{PW} marks the transition temperature between partial free cooling and mechanical cooling.

$$T^t_{TW} = T^t_{sw} - T_{ap} - T^t_{tr} \tag{46}$$

$$T^t_{PW} = T^t_{rw} - T_{ap} - T^t_{tr} \tag{47}$$

$$\text{Full free cooling mode: } T^t_{wb} \le T^t_{TW} \tag{48}$$

$$\text{Partial free cooling mode: } T^t_{TW} \le T^t_{wb} \le T^t_{PW} \tag{49}$$

$$\text{Mechanical cooling mode: } T^t_{PW} \le T^t_{wb} \tag{50}$$

Where T^t_{wb} is outdoor wet bulb temperature. T^t_{sw} is supply chilled water temperature and T^t_{rw} is return chilled water temperature. T_{ap} is approaching temperature of the cooling tower. T^t_{tr} is temperature rise of heat exchanger. $Q^t_{cool,FC}$ can be obtained by Eq. (51)or (52). T^t_{ct} is output water temperature of cooling tower, and $T^t_{ct} = T^t_{wb} + T_{ap}$.

$$Q^t_{cool,FC} = \varphi_{cool,FC} \times CAP_{FC} \quad \text{(Full free cooling mode)} \tag{51}$$

$$Q^t_{cool,FC} = \varphi_{cool,FC} \times MC \times \left(T^t_{rw} - T^t_{ct}\right) \quad \text{(Partial free cooling mode)} \tag{52}$$

For cooling storage system， $Q_{cool,CS,dis}^{t}$ is cooling supply of cooling storage and Eq.(53) – (55) are used to guarantee that system could only charge or discharge. Eq. (56) and (57) describes the remaining cooling capacity in the cooling storage tank [68-69].

$$0 \le Q_{cool,CS,ch}^{t} \le \sigma_{cool,CS,ch}^{t} \times CAP_{CS} \tag{53}$$

$$0 \le Q_{cool,CS,dis}^{t} \le \sigma_{cool,CS,dis}^{t} \times CAP_{CS} \tag{54}$$

$$\sigma_{cool,CS,ch}^{t} + \sigma_{cool,CS,dis}^{t} \le 1 \tag{55}$$

$$Q_{cool,CS}^{t} = Q_{cool,CS}^{t-1} + \varphi_{cool,CS,ch} \times Q_{cool,CS,ch}^{t} - Q_{cool,CS,dis}^{t} \Big/ \varphi_{cool,CS,dis} \tag{56}$$

$$Q_{cool,CS}^{t} \le CAP_{CS} \tag{57}$$

4. Case study

A case study was conducted in a data center located in Shanghai, which includes four IT buildings and one office building, with a total area of 65000 m². The design includes approximately 3600 racks per IT building, each consuming 5 kW of power. Currently, only one IT building is in operational with a total of 10,008 servers on the racks. This IT building is equipped with six sets of gas internal engines, absorption chillers, and electrical chillers; however, due to the low usage rate, only one set of the equipment is operational. Technical and economic parameters for energy equipment are listed in Table 1. The carbon emission factors of natural gas and grid are set at 0.18 kg/kWh and 0.77 kg/kWh, respectively, according to the East China grid's standards. Carbon tax is 0.05 CHY/kg. Prices for natural gas and electricity based on time of use, are provided in Table 2.

Table 1. Technical and economic parameters of the energy equipment

Items	Parameters	Values	Units
Gas internal engine	Capacity	2000	kW
	Power efficiency	0.42	-
	Heating efficiency	0.45	-
	Maintenance cost	0.035	CNY /kWh
Absorption chiller	Capacity	2018	kW
	Efficiency	0.9	-
	Maintenance cost	0.014	CNY /kWh
Electrical chiller	Capacity	2095	kW
	COP	5	-
	Maintenance cost	0.014	CNY /kWh
Cooling storage	Capacity	1000	kW
	Stored efficiency	0.9	-
	Charge efficiency	0.9	-
	Discharge efficiency	0.9	-
	Maintenance cost	0.007	CNY /kWh
Photovoltaic	Capacity	13500	m^2
	Maintenance cost	0.05	CNY /kWh
Free cooling	Capacity	1750	kW
	Maintenance cost	0.012	CNY /kWh

Table 2. Time of use electricity price and natural gas price

Items	Units	Peak	Flat	Valley
		10 - 15, 19 - 21	07 - 09, 16 - 18, 22 - 24	01 - 06
Electricity	CNY /kWh	1.257	0.787	0.299
Natural gas	CNY /kWh	0.343	0.343	0.343

The case study includes workload transfer and energy dispatch in summer and in winter. Figure 5 presents the solar radiation intensity data. In winter, the full free cooling mode satisfies all cooling demands, whereas the mechanical cooling mode, where free cooling is non-contributory, is similar to summer conditions. Therefore, the focus in winter shifts to studying partial free cooling mode, with wet bulb temperatures given in Figure 5. The threshold temperature T_{TW}^t and T_{PW}^t are 5 ℃ and 11 ℃, respectively. The servers are the most important IT equipment in a data center. The servers' characteristics include a maximum power of 274W and an idle power consumption of 55W. The power of other IT auxiliary equipment is 20kW. Workload processing speed of each server is 30 requests/s which means one server can process 30 workloads requests per second [60, 70-71], and the data center's waiting queue capacity is 500M requests. The server's maximum utilization is limited to 90%. In addition, the modeling and optimization of this study are performed in MATLAB R2020a, and Groubi solver

is employed for solving optimization. All processes are conducted on a computer with an Intel-i9 processor and 16 GB of RAM.

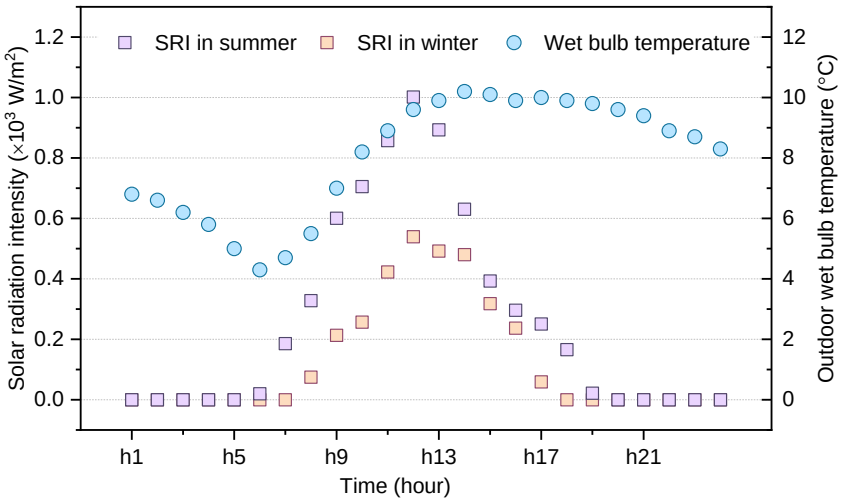


Figure 5. SRI and wet bulb temperature of the case study.

5. Results and discussion

5.1. Trade-off between workload transfer and energy dispatch

The results of integrated optimization between workload transfer and energy dispatch are shown in Figure 6. This illustration reveals a clear trade-off between the ADR and the total operational cost associated with managing the energy system. Typically, transferring workloads reduces the energy system's total expenses at the cost of increasing the ADR. During summer operational conditions, the ADR reaches a peak of 0.2597, while the operational costs are cut from 13332 CNY to 12315 CNY. In winter, the ADR reaches a maximum of 0.2028, with the energy system's operational costs ranging between 15371 CNY and 14939 CNY showing a less pronounced variation in ADR compared to the summer. The points labeled W1 and S1 in the figure denote the lowest operational costs for the energy system in winter and summer respectively, without any workload transfer. Conversely, the points W3 and S3 illustrate the minimal operational costs are achieved through workload transfer. It is noteworthy that the workload's ADR increases from W2 to W3 and from S2 to S3, whereas the energy system's operational cost remains unchanged. We refer to the workload transfer of this part as "useless transfer", suggesting that workload transfers do not always bring benefits for the energy system.

Therefore, the data points spanning from W2 to W3 and from S2 to S3 will be excluded from subsequent analyses.

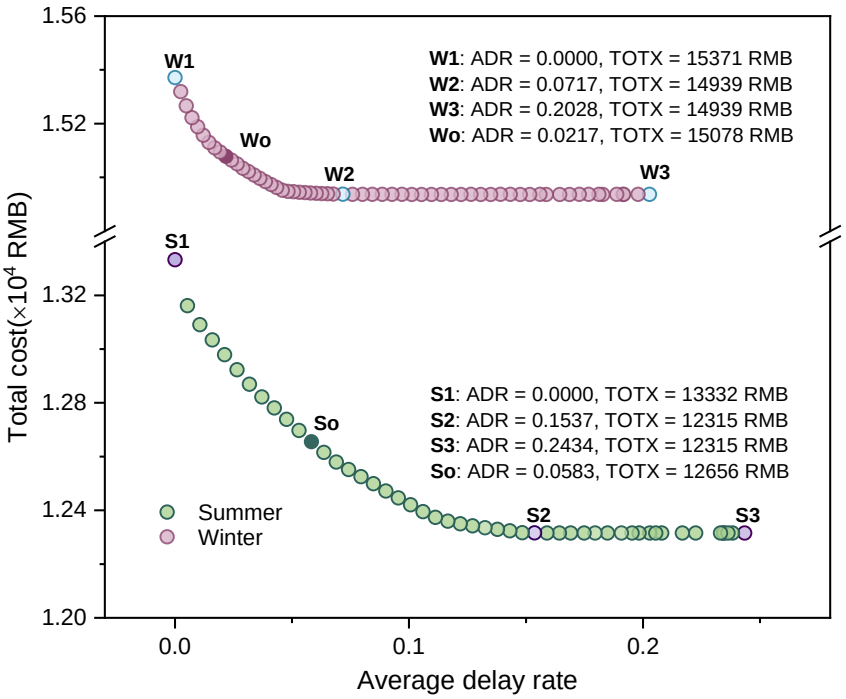


Figure 6. Pareto frontier for cross-sectoral optimization.

The workloads arriving at date center is shown in Figure 7 (a) and (b). Delay-sensitive workloads come from a real-world dataset [24, 72]. Delay-tolerant workloads are generated based on existing literature [24]. The workloads used this study are listed in Table 3. There are 5 types of workloads in Figure 7 (a) namely D1-D5. “Dn” means this type of workloads can be delayed for n hours. The value is calculated by workload’s arrival time and deadline. If a workload arrives at 8 am and its deadline is 11 am, it can be delayed for 3 hours, making it a D3 workload. The maximum value of delay tolerant workloads is 241M at h9. The transfer of workloads is influenced by four critical factors: (1) the affordability of the data center’s IT equipment; (2) the configuration capacity of the energy system’s equipment, (3) the total operating cost of the energy system, and (4) the ADR of workloads. As depicted in Figure 7(c), both the affordability of IT equipment and the configuration capacity of energy system surpass the demands of the current workloads. Hence, in the current case, operating costs and ADR of workloads are the main constraints on workload transfer.

Figure 7 (d) - (f) show the workload profiles. Figure 7 (d) presents the workload profile with no transfer. Figure 7 (e) and (f) depict workload profiles under different transfer scenarios. Typically, the first few hours of the day (such as 1 am to 4 am in this study) may receive workloads transferred from the previous day, resulting in higher workloads at those times compared to the original workloads. However, in these figures, only the workload transfers of the current day are considered, and transfers from the previous day are not taken into account. Therefore, the workloads from 1 am to 4 am in Figure 7 (e) and (f) do not increase.

When the workloads are transferred at its maximum transfer capacity — specifically, transferring all delay tolerant workloads to the latest possible time slot within their deadlines, the ADR is 0.5483, as depicted in Figure 7(f). This ADR maximum notably surpasses the peak ADRs recorded in Figure 6, which are at 0.2434 for summer and 0.2028 for winter. ADR values exceeding 0.2434 and 0.2028 are not captured by the Pareto frontier, which suggests that the condition requiring at least one of the two objectives becomes smaller cannot be satisfied. Therefore, it can be concluded that extending workload transfers beyond points W3 and S3 will increase the total operating cost of the energy system, demonstrating that workload transfers do not always bring benefits for energy system and that the impact of it could even be negative. Combined with the results in Figure 6, as the workload transfer increases, the total cost of the energy system first decreases, then remains unchanged, and eventually increases.

The points WO and SO in Figure 6 represent the optimal decision-making points obtained by TOPSIS and the workloads after being transferred at these two points are shown in Figure 7 (e). The segments W1-Wo/S1-So and Wo-W2/So-S2 highlight where the ADR predominantly influences and where the total operating cost takes precedence, respectively. The ADR variation from S1 to So is 0.0583, accounting for 37.93% of total variation, whereas cost decrease from S1 to So is 676 CNY, accounting for 66.47% of total cost decrease. Similarly, ADR variation from W1 to Wo is 0.0217, accounting for 30.26%, whereas cost decrease is 293 CNY, accounting for 68.14%. Regions where ADR is more dominant see more significant cost savings from increases in ADR. This suggests a non-linear impact of workload transfer on the cost saving of the energy system, with diminishing benefits as workload transfer increases.

Figure 7 (d) and (e) depict workloads profiles at of points W1/S1 and Wo/So, with peak-to-mean ratios of 1.56 in Figure 7 (d), 2.45 (Summer)/1.67 (Winter) in Figure 6(e), and in Figure 7(f) its 1.93. These figures illustrate an increase in workload fluctuations as the increase of workload transfer, which is related to the driving force of the transfer, which will be discussed in section 5.2.2.

In this study, in order to emphasize the differences in the impact of workload transfer between summer and winter, the case for winter and summer are calculated separately. It is noticeable that there is a significant difference in the optimal ADR between summer and winter. The optimal ADR for summer is relatively high when considered for winter conditions. Future research is needed to analyze the workloads throughout the year in order to determine the overall optimal point for the entire year.

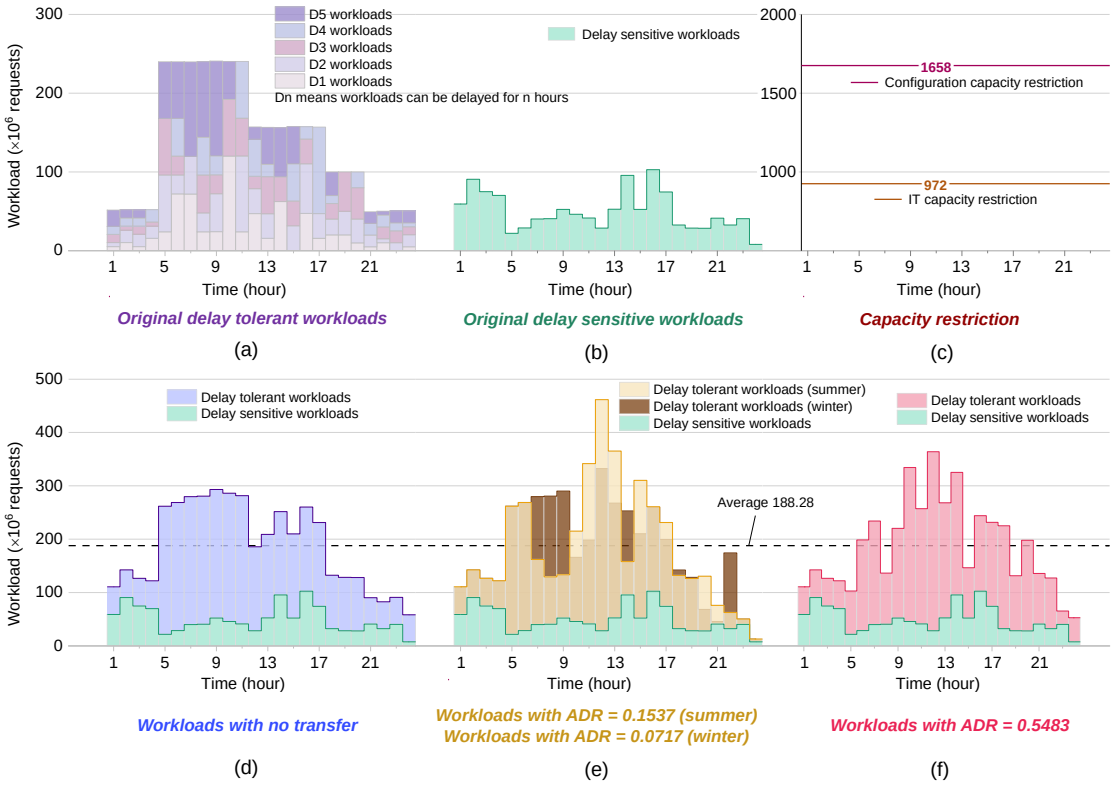


Figure 7. Workloads (a) original delay tolerant workloads; (b) original delay sensitive workloads; (c) capacity restriction of workload transfer; (d) workloads with no transfer; (e) workloads of point W3 and S3; (f) workloads at maximum ADR within transfer capability.

Table 3. Workloads used in this paper. ($\times 10^6$ requests)

Time	Delay-sensitive	Delay-tolerant				
		D1	D2	D3	D4	D5
1	59	5	5	10	10	21
2	91	10	16	5	10	10
3	75	5	16	10	10	10
4	70	16	16	5	16	0
5	22	24	72	72	0	72
6	29	72	24	24	48	72
7	40	72	48	0	0	120
8	41	24	24	48	48	96
9	52	24	48	24	24	120
10	46	120	0	72	0	48
11	42	24	96	48	72	0
12	29	47	31	16	47	16
13	53	16	31	47	16	47
14	96	62	0	31	0	62
15	53	0	32	32	47	47
16	103	47	63	31	16	0
17	74	16	31	0	110	0
18	33	20	20	20	10	30
19	29	20	30	50	0	0
20	29	10	30	40	20	0
21	41	5	15	0	15	15
22	33	10	5	15	15	5
23	41	0	10	15	10	15
24	8	5	15	10	5	15

The energy demand of a data center is closely linked to the transfer of workloads. As the electrical demand in data centers is not significantly different from the cooling demand, this discussion will primarily focus on the electrical demand. The variation of electrical demand is shown in Figure 8. The horizontal axis represents the time of day, and the vertical axis represents 'Position on the Pareto Frontier'. In this paper, the axis 'Position on the Pareto Frontier' is related to the data in Figure 6. In the summer data plot, the 'Position on the Pareto Frontier' values from 1 to 30 correspond sequentially to the 30 points between S1 and S2 in Figure 6. Similarly, in the winter data plot, the 'Position on the Pareto Frontier' values from 1 to 30 correspond sequentially to the 30 points between W1 and W2 in Figure 6. From position 1 to position 30, the situation corresponds to an increase in ADR while the total operational costs of the data center decrease.

The observed variation in electrical demand, both in terms of duration and range, is higher in summer compared to winter. Specifically, the peak demand reaches 1389 kWh in winter and 1686 kWh in summer, with both peaks occurring at h12. During daylight hours, the utilization

of clean PV energy facilitates the transfer of workloads in both seasons. This transfer increases the use of clean electricity, reducing reliance on the grid power, which in turn lowers operational costs. The lowest point of the electrical demand is 776 kWh in both seasons at h24. The electricity price at h1 is significantly lower than the electricity price at h24, causing a large number of workloads to transfer out of h24, resulting in a decrease in the electrical demand at h24.

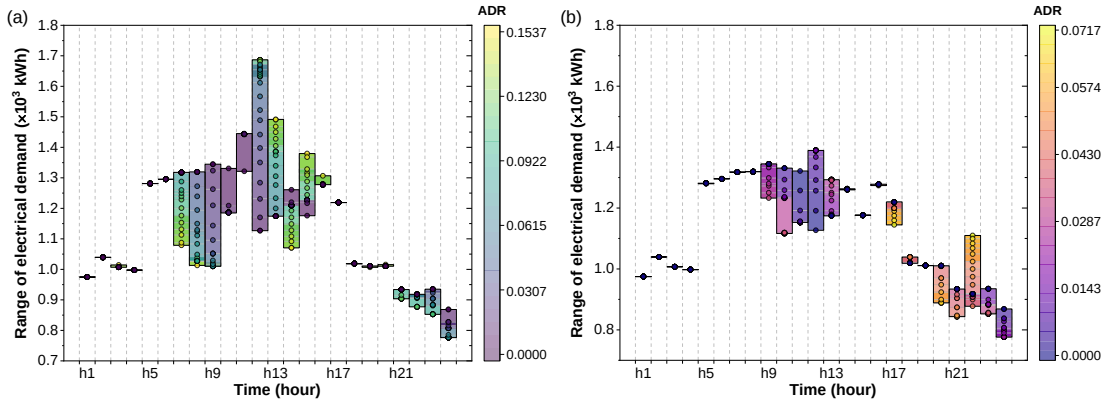


Figure 8. Range of electrical demand (a) in summer and (b) in winter.

Figure 8. presents the cost analysis of the energy system. In summer, as illustrated in Figure 9 (a) - (d), the expense associated with gas consumption remains largely stable, with a slight increase in its cost proportion. Conversely, both the cost and proportion of electricity consumption exhibit a decrease, alongside a reduction in the unit cost of electricity. This suggests that workloads transfer tends to favor periods of lower electricity price. There is an increment in total maintenance costs, attributed primarily to increased utilization of PV. The cost of carbon emission has slightly decreased. Notably, the carbon emissions from natural gas have seen no significant increase, whereas those from grid electricity consumption have decreased. However, given the relatively low carbon tax, the decrease in the cost carbon emissions is not much. The significant increase in PV usage, with curtailment reduced from 7.80% to 0, and a reduction in grid electricity consumption, primarily contribute to the overall decrease in carbon emissions of the energy system. In winter, as depicted in Figure 9(e)-(h), the trends in maintenance and carbon emission costs are similar to those observed in summer. The cost of gas and electricity show non-monotonic variations, initially due to the rise in PV

utilization and the transfer of workloads to periods with lower electricity price. Subsequently, workloads are transferred to periods when gas internal engine is operational, resulting in an increase usage of gas internal engine and a decrease in grid electricity. Finally, the workloads are further transferred, leading to a rise in the usage of free cooling. This transfer reduces the need of absorption chiller, so the use of gas internal engine decreases.

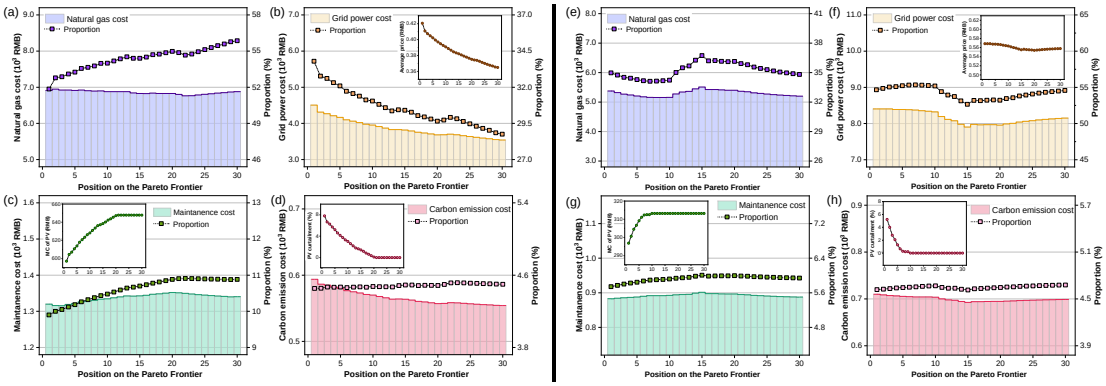


Figure 9. Cost of energy system in (a) - (d) summer and (e) - (h) winter.

These results indicate that optimizing either the IT sector or the energy sector independently can adversely impact the performance of the other. This is primarily due to the coupling relationship of energy and data, which complicates the achievement of optimal solutions in both sectors without a cooperative strategy. Therefore, through the integration of workload transfer and energy dispatch, the potential for cost and carbon reduction in the data center can be fully explored in an economical and sustainable way.

5.2. Analysis of workload transfer

5.2.1. The workload transferred out of each time slot

The workloads arriving at the tth time slot can be transferred out of their initial time slot, with the set of feasible transfer time slots represented by Φ_k^t. Figure 10 illustrates the results of workload transferred out of each time slot in summer and winter, respectively. In this figure, the horizontal axis denotes the arriving time of workloads, while the vertical axis indicates the position on the Pareto frontier. The size of the bubbles in the figure reflects the proportion of the transferred workloads to the total delay tolerant workloads in that time slot, and the color

gradient represents the amount of workload transferred out. The workload transfer in summer mainly occurs in h7-h11 and h21-h24; in winter in h10-h12 and h20-h24. Given that the minimum transfer time is one time slot, workloads with a delay interval of one time slot (D1) are ineligible for transfer. Therefore, subtracting the workloads with delay interval of one time slot from delay tolerant workloads is what can be moved, so the maximum proportion value in Figure 10 is 0.9 instead of 1.

In Figure 10, it can be seen that the transfer of workloads increases in a stepped pattern from point1 to point30. This is closely related to the structure of the workloads, encompassing both inherent characteristics and the heterogeneity in their composition. On the one hand, the inherent characteristics of the workload itself, specifically the length of the delayable interval, results in different delay rate for a given duration of response time. As listed in Table 4, the delay rate caused by the waiting time, which plays a dominant role in total delay rate, shows a stepwise increase in value. Having the same waiting time, for instance, 2 time slots, results in varying delay rates: 0.67 for D3, 0.50 for D4, and 0.4 for D5. On the other hand, the composition of arriving workloads at the data center introduces additional variability. At each timeslot, the composition of the workload arriving at the data center varies. For instance, at h20, the absence of D5 workloads (see Figure 7(a)) means no workload being waiting for 4 time slots. These variations in characteristics and composition influence the amount of workload transfer, which subsequently impacts the advantages derived from workload transfer.

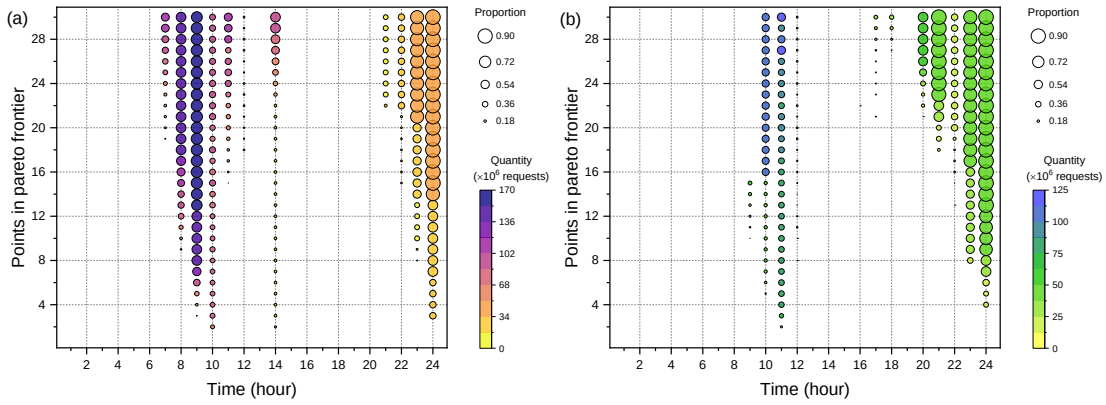


Figure 10. The results of workload transferred out of each time slot (a) in summer and (b) in winter.

Table 4. Delay caused by waiting time

Workload type	Waiting time	Delay rate	Rank
D2	1	0.50	5
D3	1	0.33	3
	2	0.67	8
D4	1	0.25	2
	2	0.50	5
	3	0.75	9
D5	1	0.20	1
	2	0.40	4
	3	0.60	7
	4	0.80	10

For different types of workloads at the same time slot. In the context of identical benefits derived from transferring workloads, the sequence of these transfers is primarily determined by the delay rates attributed to waiting times, as detailed by the “Rank” column in Table 4. Take the workloads from h21 to h24 in summer as an example to illustrate. These workloads are all transferred due to the difference in electricity prices, arriving at h1 of the following day. Figure 11 illustrates the sequence in which the workloads are transferred during the increase of the ADR. The x-axis represents positions on the Pareto frontier, with ADR increasing from point1 to point30, while the y-axis shows the amount of delay tolerant workloads calculated at this time slot. Pareto point1 depicts the scenario from h21 to h24 without any transfer. The initial transfer involves workloads D4 and D5 of h24, followed by D3, and then D5 of h23. They are transferred according to their rank order, with D5 from h21 being the last to be transferred.

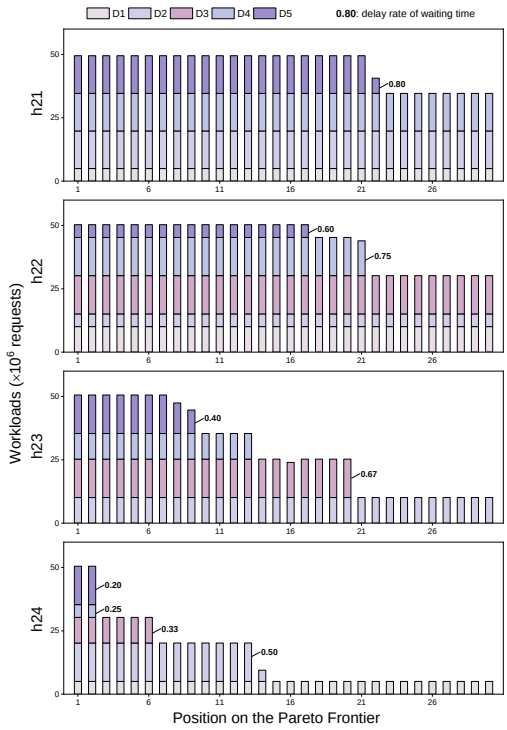


Figure 11. Delay tolerant workloads calculated at h21 - h24 after workload transfer.

There is also a sequence of transfer at different time slots. Taking summer in Figure 10 (a) as an example, h10 and h14 are the first two time slots starting transfer at point2 in Pareto frontier. At h9, h8, and h7, transfer starts at points3, points9, and points19, respectively. The latest time slot is h21 who starts at point22. This is related to the timing of the driving force of transfer (which will be analyzed in the next section). For instance, at h11, there is PV curtailment and at h15, the gas internal engine set starts, allowing the workloads of h10 and h14 to transfer to h11 and h15 respectively. At h9, h8, and h7, the driving force adjacent to them is the excess PV power at h11. The delay rate of workload transferred from the three time slots to h11 increase sequentially, leading to a corresponding sequential postponement of the transfer initiation points on the Pareto frontier.

5.2.2. The workload transferred into each time slot

Workloads are transferred upon the realization of benefits, which come from three aspects in this study: (i) the difference electricity prices, as shown at h1 of the next day in Figure 12 (a) and (b); (ii) the excess renewable resources, noted at h11-h13 in Figure 12(a) and (b); at h22 in

Figure 12 (b); (iii) The optimal operation of the energy system, as shown at h15 in Figure 12 (a), and at h18 in Figure 12 (b)

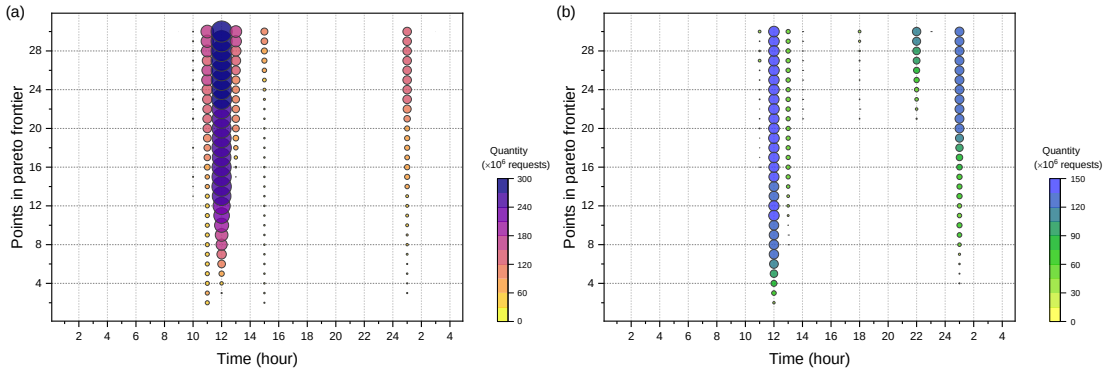


Figure 12. The results of workload transferred into each time slot (a) in summer and (b) in winter.

One of the driving forces of workload transfer is the revenue from the difference in electricity prices. As stated in the previous section, electricity prices are lower at h1 of the next day, and workloads from h21 to h24 will be moved to h1 to reduce cost of purchasing electricity. These are part of workload transfer caused by different electricity prices. The average electricity decrease is 13.24% in summer and 1.88% in winter, as shown in Figure 9 (b) and (f).

The second driving force behind workload transfer is the excess renewable energy resources. Here, excess renewable energy refers to the unused renewable energy when no workload transfer occurs. Part of it comes from PV curtailment. For instance, in summer case, original workloads with no transfer leave unused PV power at h11, h12 and h13, which will become the target time slots for workload transfer. PV utilization reaches 100% sequentially at h11, h12, and h13, notably during intervals from point1 to point 2 (for h11), from point 2 to 14 (for h12), and from point 14 to 21(for h13), as depicted in Figure 13 (a). The workloads allocated to these three time slots predominantly originate from those at time slots of h7 - h10. The nearest time slot with PV surplus is h11, to which the workloads will be first moved. After all the PV power at h11 is used up, workloads will start to be moved to h12, and finally to h13. Therefore, the PV utilization rate from h11 to h13 increase to 100% sequentially. In winter, PV driven workload transfer is similar and will not be detailed again. In addition, another part of driving force is the extra free cooling in winter. The use of free cooling increases by 1.10%.

There is surplus free cooling during the periods of h22~h24 and h1~h9. At h18, h20, and h21, there is workload transfer to the h22. However, since there were no D4 and D5 workloads at h19, there is no workload transfer to h22 from h19.

The third driving force is the optimization of energy system operation. For example, in the summer case, the workload of h14 is transferred to h15. The variation of gas internal engine from point 1 to point 30 in Pareto frontier can be divided into three stages, as shown by the blue scatters in Figure 13 (b). Firstly, a small number of workloads was transferred from h14 to h15, which resulted in a decrease in the electricity purchase of the h14 to zero, as shown by the purple scatters in Figure 13 (b). Then, within the range of increasing PV usage, there was no workload transfer from h14, and the output of gas internal engine remains unchanged until the surplus PV is fully utilized. Afterwards, the workload of h14 begins to move to h15, and the output of gas internal engine at h15 starts to increase. Meanwhile, the discharge of cooling storage of h14 begins to decrease (green scatters in Figure 13 (b)), and the output of electrical chiller begins to increase (red scatters in Figure 13 (b)) until all movable workloads of h14 have been fully transferred. Through the transfer of workloads, h14's electrical demand and 94.02% of its cooling demand are all supplied by PV power, while the remaining 5.98% of its cooling demand is supplied by the discharge of cooling storage. It not only reduced the amount of electricity purchased from the grid but also reduced the cooling storage at the pre sequence time, further reducing operating costs.

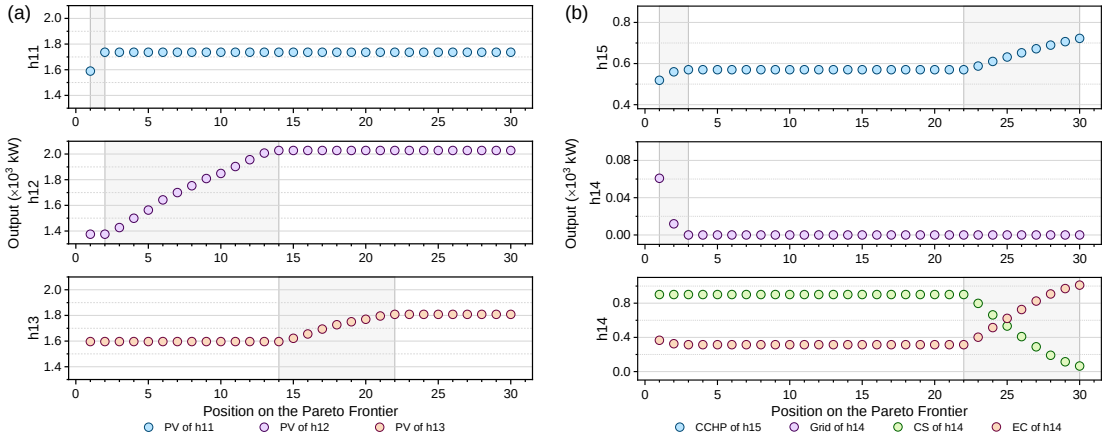


Figure 13. The variation of (a) PV output in h11 - 13 and (b) energy equipment output in h14 -15

The transfer tendencies of different types of workloads are also different. Figure 14 is the average delay rate of each type of workload in summer (the total delay rate of a given type of workload divided by its the total amount) and the percentage of transferred workloads. Both of these decrease from D5 to D2. Specifically, the average delay rates for D5 to D2 workloads stand at 0.4037, 0.1632, 0.1299, and 0.0130, respectively. The proportion of untransferred workloads for D5 to D2 is 37.5%. 68.3%, 72.9% and 97.4%, respectively. Figure 15 shows the proportion of the average delay rate of each type of workloads in the total delay rate. In Figure 14, the average delay rate of D5 increases, but in Figure 15, the proportion of D5 in ADR shows a downward trend. The average delay rates of D4 and D3 are also on an upward trend overall, but in Figure 15, the proportions of D4 and D3 in the total delay rate are not much different. D2 has the lowest average delay rate and minimal variation overall.

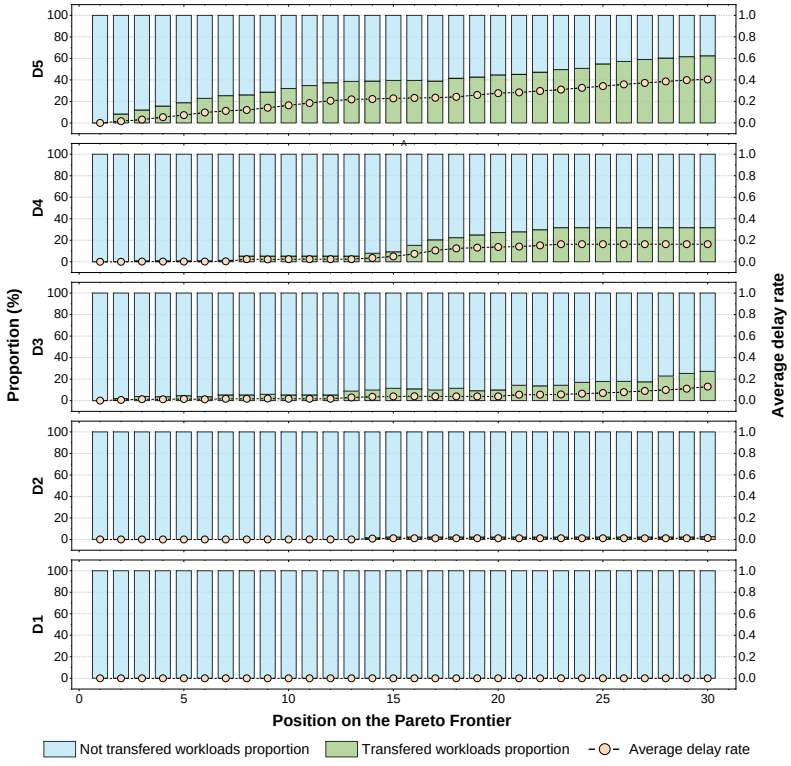


Figure 14. The transferred proportion and average delay rate of each type of workload.

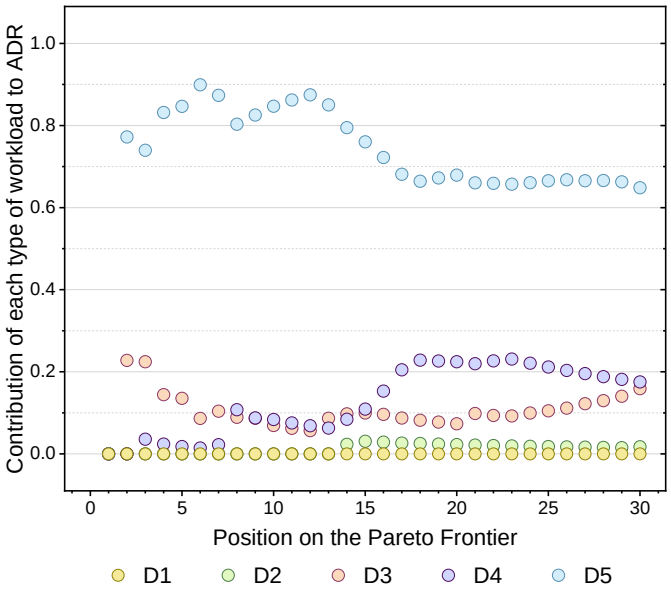


Figure 15. Contribution of each type of workload to ADR

5.3. Analysis of energy dispatch

Figure 16 shows the electrical demand and the output of each energy equipment. Figure 16 (a) - (c) shows the results in summer, where the grid supplies power from h1 to h10, while the gas internal engine provides power from h15 to h24. The remaining periods are covered by PV output. It is noted that the transfer of workloads does not influence the activation times of these energy equipment. In Figure 15 (a), the peak load is observed at h10, whereas, in Figures 15 (b) and (c), the peak occurs at h12. In scenarios without workload transfers, the demand is impacted by the workload arrival rate. However, with workload transfers, the demand is also shaped by the driving forces. At h12, PV output is substantial, leading to a concentration of workloads at this time. And also due to higher PV utilization, the electricity usage of the electrical chiller also increases. In winter, PV generates less power and is not the main driving force for workload transfer, so load fluctuations in winter are smaller. The gas internal engine operates over a longer duration from h10 to h21. The amount of electricity import from grid is largely affected by the time-of-use electricity price; notably, during the peak price intervals of h10 to h15 and h19 to h21, the grid's electricity purchase drops to zero. Furthermore, free cooling in winter markedly diminishes the electrical chiller's output compared to summer, with the electrical chiller's power consumption nearing zero.

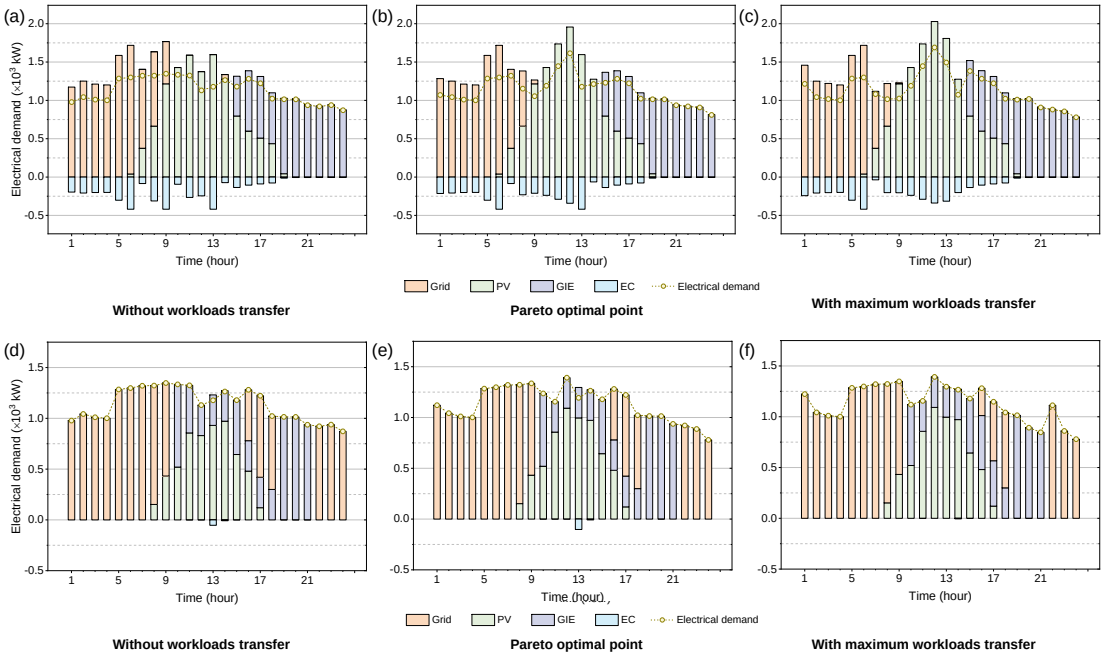


Figure 16. Electrical balance in summer, (a) point S1; (b) point So; (c) point S2 and in winter, (d) point W1; (e) point Wo; (f) point W2.

Figure 17 depicts the energy balance of the cooling system. In summer, the gas internal engine operates between h15 and h24. During these hours, absorption chiller serves as the primary cooling equipment. Outside this period, the electrical chiller functions as the primary cooling device. In winter, the primary cooling method is free cooling, with the absorption chiller playing a minor role, and the electrical chiller being largely inactive. As depicted in Figure 17 (d) to (f), with the transfer of workloads, there is an increase in the utilization of free cooling, resulting in a decrease in the use of absorption chiller. Generally, the transfer of workloads does not impact the start-stop schedule of equipment but primarily influences their output. The cooling storage is notably affected by workload transfers. In summer, workload transfers eliminate the need for discharges at h10 and h14. Similarly, in winter, it leads a decrease in the use of cooling storage, although the impact is less pronounced.

The data centers that implement workload transfers may require smaller capacity for cooling storage. Figure 18 shows the variation of cooling storage use intensity in summer. It is observed that cooling storage usage substantially decreases with the implementation of

workload transfers. Cooling storage equipment serves a crucial role in demand regulation. From this perspective, the operational function of cooling storage parallels that of workload transfer. Consequently, an increase in workload transfers results in reduced utilization of cooling storage. Hence, for data centers implementing workload transfers, there may be a necessity to reevaluate the capacity of cooling storage equipment.

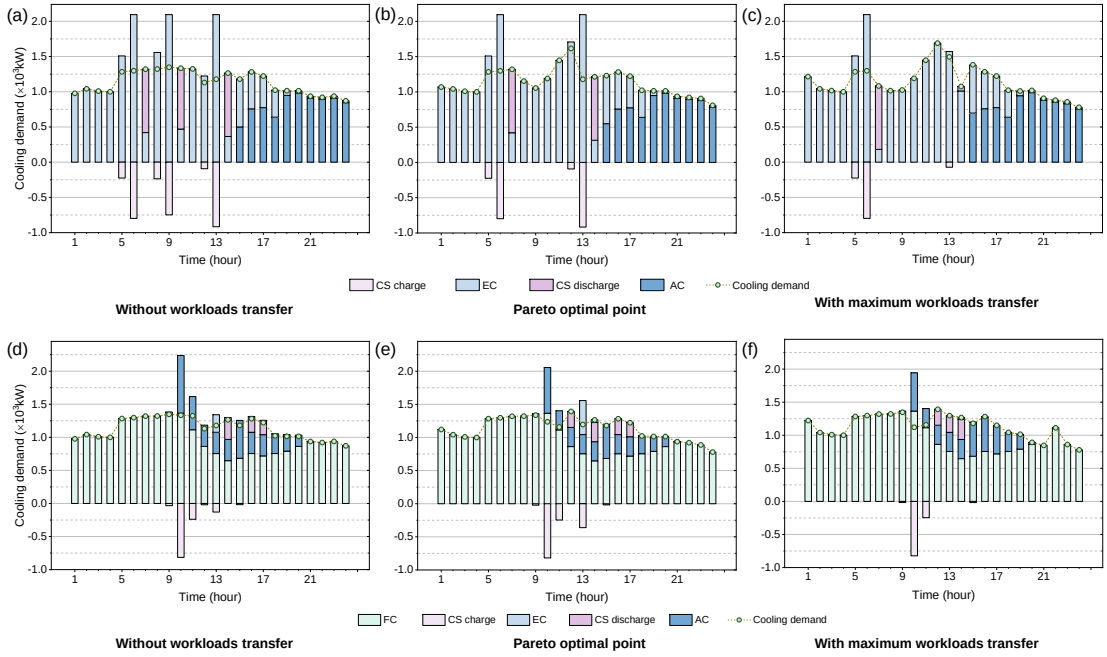


Figure 17. Cooling balance in summer, (a) point S1; (b) point So; (c) point S2 and in winter, (d) point W1; (e) point Wo; (f) point W2.

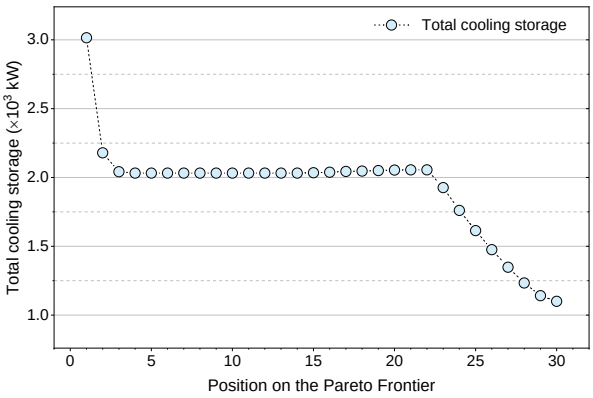


Figure 18. Total cooling storage in summer

Figure 19 illustrates the workload transfer and variations in operating costs of Pareto optimal point. From the figure, it is evident that under summer conditions, significant workload transfers occur during periods of surplus PV energy, particularly around h11 and h12. These transfers enhance the utilization of PV energy, thereby reducing reliance on grid electricity, lowering operating costs, and decreasing carbon emissions. Under winter conditions, the limited surplus PV energy leads to less workload transfer. In both summer and winter, the lower electricity price in the early morning hours result in nighttime workloads being transferred to h1 of the following day. As shown in the figure, the operating costs from h23 to h24 are reduced.

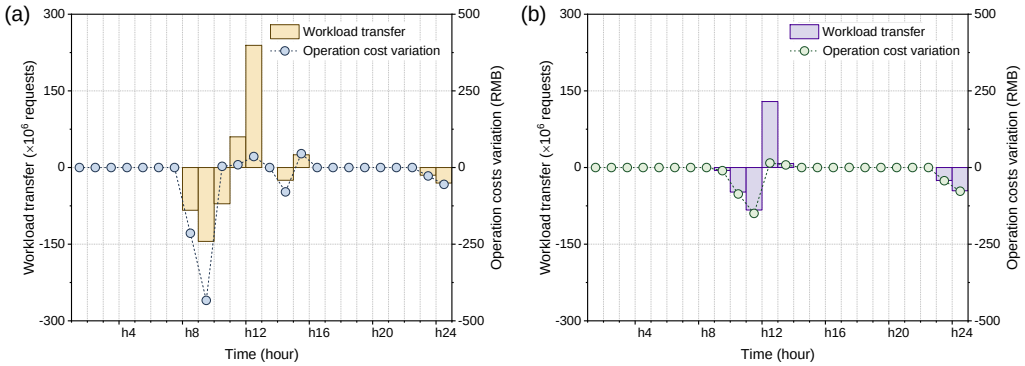


Figure 19. Workload transfer and operating costs variation of Pareto optimal point (a) in summer and (b) in winter.

6. Conclusions

To thoroughly explore the potential for sustainable development in data centers and ensure their healthy growth, this paper analyzes the bidirectional coupling relationship between energy and data in data centers. Based on this, an integrated management framework combining a workload transfer module and an energy system optimization module is proposed to achieve cross-sector management of the IT and energy sectors. Furthermore, the proposed methods are applied in a case study of a data center located in Shanghai. The main conclusions are as follows.

1. A clear trade-off exists between ADR and the operational costs of energy systems. Specifically, transferring workloads could reduce energy system costs by 7.63% in summer and 2.81% in winter, but it also increases the ADR by 0.2434 in summer and 0.2028 in

winter. An excessive focus on cost reduction could raise the ADR, negatively affecting user satisfaction. Moreover, the impact of workload transfer on the operational costs of energy systems is not uniformly positive. According to the results of a case study, within the full transferrable range, the total operational costs initially decrease with the transfer of workloads, subsequently stabilize, and ultimately increase.

2. The workloads structure significantly influences the workload transferred out of each time slot. The number of workloads transferred out exhibits a stepwise increase. Typically, workloads with the lowest delay rates due to waiting times are prioritized for transfer, with the largest ones being transferred last. The time slots of workload transferred into are affected by the driving force, which, as evidenced by case study, mainly include difference in electricity prices, the additional renewable energy resources, and the optimization of energy system operation. The transfer of workloads results in a significant decrease in the average electricity price, with reductions of 13.24% in summer and 1.88% in winter. This also leads to an increase in PV utilization by 7.80% in summer and 5.23% in winter, and also makes free cooling usage increase in winter by 1.10%.
3. The transfer of workloads primarily influences the output of energy equipment and hardly changes the startup schedule of the equipment. The workload transfer has a greater impact on the cooling storage system. Since both workload transfer and cooling storage systems have the function of demand shift, an increase in transfer results in a decreased use of the cooling storage system. Compared with no transfer results, the usage of the cooling storage system drops by up to 2000kW in summer.

In summary, this study proposes an integrated management framework for workload transfer and energy dispatch, considering the coupling relationship between them in data centers. The proposed method provides a new tool for synergistic management in data centers, presenting managers with new insights based on the results of the case study.

However, the current research has some limitations. It primarily focuses on the joint management of workload transfer and energy dispatch for data centers on a day-ahead basis. Building on this foundation, future studies should investigate real-time scheduling of workload

and energy. We aim to obtain actual operational data from data centers and apply machine learning methods to develop models for real-time workload and energy scheduling. Additionally, this study did not address the interaction between multiple data centers. In the future, we plan to explore workload scheduling for multiple data centers using blockchain technology to fully harness the potential of spatial workload scheduling in energy conservation and emission reduction.

Acknowledgments

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